Choosing your Market: Information and Competition

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Motivation

Matching markets are ubiquitous, and well-studied.

- Typically, the set of agents in a market is taken to be exogenous, and then frictions within a market are studied.

Often, this set of agents is endogenous. Agents make choices that sort them into different (sub-)markets.

- What ***kind*** of education/product/research do I pursue/produce/undertake?

How do market frictions affect this choice of market?

- When should we expect these choices to be efficient?
- If they are inefficient, what form will the inefficiency take?

Model Overview

"Sorting" **agents** with heterogeneous quality choose which market to enter.

Within a market, agents match one-to-one with heterogeneous "stationary" agents: **positions**.

- Surplus is created, depending on the matched pair.
- This surplus is divided between the agents in the pair.

I explore the impact of two frictions:

(i) **Information frictions**: whether agents know their quality when they choose their market.

(ii) **Competition frictions**: whether agents compete for matches (TU vs. NTU).

Main Result

	No Quality Uncertainty	Quality Uncertainty
Competitive Markets	No Inefficiency	No Inefficiency
Noncompetitive Markets	No Inefficiency	**“Allocative” Inefficiency**

Applications

Apply the framework to three applications:

	Sub-Markets	Agents (Quality)	Positions (Attribute)
Workers	Sector	Workers (Productivity)	Firms (Productivity)
Innovation	Field	Researchers (Success Time)	Innovation (Value)
Production	Genre	Producers (Quality of Good)	Demand (Quantity)

Motivating Example

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There are two agents, and two markets, each with one stationary agent (position).

- Agents' quality are distributed according to $v_i \sim \text{Unif}[1, 2]$.
- Sector 1's position has attribute $z_1 > \frac{1}{2}$, Sector 2's has attribute $z_2 = 1 - z_1$.
- If agent i matches with the position in sector j, surplus $v_i \cdot z_j$ is created.

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- They may or may not observe qualities before this decision.

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Agents each decide to enter *one* market.

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Quality then becomes public, agents match with positions, and divide surplus.

- In competitive markets, agents propose a surplus division to the position in their market. A position accepts its best offer.
- In noncompetitive markets, matched agents receives surplus proportional to the created surplus, so the best agent is matched.

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(i) If quality is public before agents choose sectors, the equilibrium is efficient;

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- If both agents enter sector 1, the worse agent $-i$ receives 0, the better agent i receives

$$(v_i - v_{-i}) \cdot z_1$$

- Agents internalize their *displacement* of the other.
  - This leads to efficiency.

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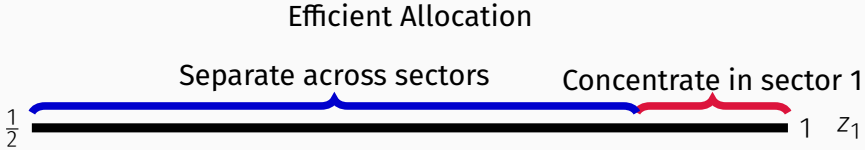


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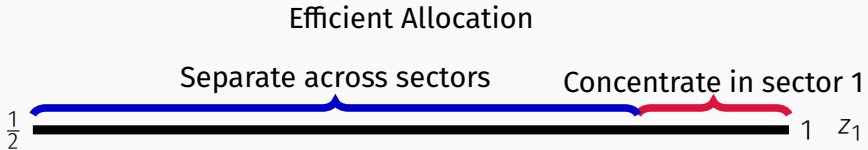
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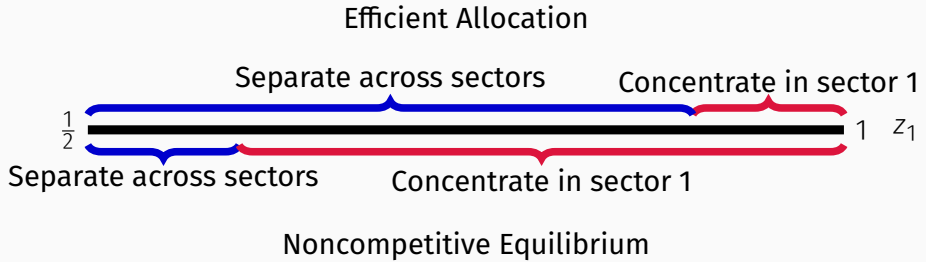
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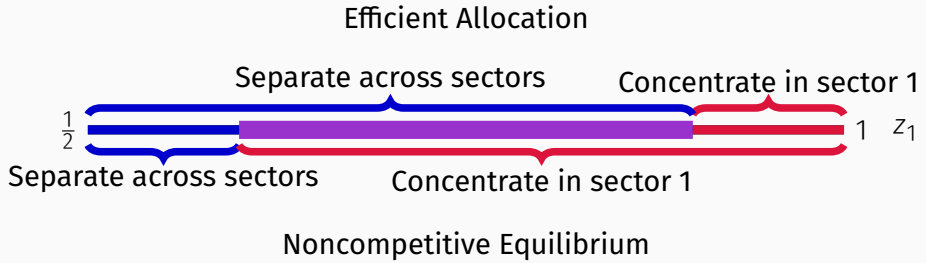


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Model

Agents and Sectors

Two-sided matching market, with one side divided into discrete markets.

- Continuum of (sorting) **agents** $i \in [0,1]$, each with quality $v_i \sim F$.
- Markets $\mathcal{M} = \{1, ..., M\}$, each with a mass of stationary agents: **positions**.
- In market m, position $j_m \in [0,1]$ has attribute $z_{j_m} \sim G_m$.

An agent commits to a market and is matched with a position in that market.

- Agent i and position j_s produce and divide surplus $\sigma(v_i, z_{j_s})$.

Assumption

(i) F is full-support on $[0, \overline{v})$ for $\overline{v} \in (0, \infty]$, with density f;

(ii) G_m on $Z_m := [0, \overline{z}_m]$ for $\overline{z}_m \in (0,1]$ has density g_s bounded from zero; and

(iii) $\sigma(v,z) \in \mathcal{C}^2$ is increasing and strictly supermodular with $\sigma(0,z) = \sigma(v,0) = 0$.

Information and Agents' Types

An agent's type is given by a signal x of their rank r within the population.

$$\mu \in \Delta\Big(\underbrace{[0,1]}_{\text{Rank}} \times \underbrace{\mathbb{R}}_{\text{Signal}} \Big)$$

Assumption

μ admits a continuous, full-support density $h(r, x)$, or perfectly reveals types.

The marginal $h_x(r)$ corresponds to agent x's beliefs about their rank r.

Extremes:

- Full information: $\mu_r = \delta_r$. Denote by μ_C.
- No information: $h_x(r) = 1$. Denote by μ_N.

Agents' Strategy

A population strategy for agents is a mapping from signals to markets:

$$\Phi : \mathbb{R} \to \Delta(\mathcal{M})$$

Agents' qualities are realized before matching occurs.

A mass of agents' realized ranks are deterministic, given their signals. Hence this strategy induces a rank-measure of agents Ψ_m in each market m:

$$d\Psi_m(r) = \int \Phi_m(x) \cdot \mu(r, dx)$$

and a distribution of qualities F_m:

$$F_m(v) \coloneqq \underbrace{\Psi_m([0, F(v)])}_{\substack{\text{Conversion from rank} \\ \text{to quality}}} + \underbrace{(1 - \Psi_m([0,1]))}_{\text{Remaining mass}}, \qquad \text{Let} \quad V_m \coloneqq \operatorname{supp}[F_m].$$

Within-Sector Matching

Given the realized ranks of agents Ψ_m, agents match with positions.

Definition

A *matching function* $\nu : Z_m \to V_m$ is a measurable function that is measure-preserving with respect to G_m, F_m.

Comment: Alternatively, could define a matching as a joint distribution over G_m, F_m. This is useful if G_m has atoms.

- For now, this distinction doesn't matter. However, if I allow G_m to have an atom at 0, then will need different notation (the results still all go through).
- I am thinking of leaving this simpler notation for a presentation, and then including the more general notation in the paper.

Division of Surplus

After agents match with positions, created surplus is divided:

Definition

- Given Ψ_m, an agent-surplus profile in m is a function $\sigma_{a,m} : V_m \to \mathbb{R}_+$.
- A market m position-surplus profile is a function $\sigma_{p,m} : Z_m \to \mathbb{R}_+$.

A matching is a collection $\left(\nu, \sigma_{a,m}, \sigma_{p,m}\right)$.

If a matching ν is feasible, $\sigma_{a,m}(\nu(z)) + \sigma_{p,m}(z) \leq \sigma(\nu(z), z)$.

There are two possible regimes for how this division is decided:

- Competitively: Agents and positions compete through surplus offers (TU); or
- Noncompetitively: There exists an exogenous division of surplus (NTU).

Division of Surplus: Competitive Markets

Competitive markets capture TU settings: agents compete for their match.

Definition

An *m competitive equilibrium* given Ψ_m is a matching $\left(\nu, \sigma_{a,m}, \sigma_{p,m}\right)$ such that

$$\sigma_{p,m}(z) = \sigma(\nu(z), z) - \sigma_{a,m}(\nu(z)) \geq \sigma(v, z) - \sigma_{a,m}(v), \quad \text{for all } z, v \quad \text{(Stab)}$$

$$\sigma_{a,m}(v) \geq 0, \quad \text{for all } v \quad (\text{IR}_a)$$

$$\sigma_{p,m}(z) \geq 0, \quad \text{for all } z \quad (\text{IR}_p)$$

Lemma

Given Ψ_m, any sector m competitive equilibrium features positive assortative matching (PAM): $\nu(z) = F_m^{-1}(G_m(z))$.

Division of Surplus: Noncompetitive Markets

Noncompetitive markets: an exogenous (market-wide) division rule: NTU.

Definition

A division of surplus $\sigma_a, \sigma_p : \mathbb{R}^2_+ \to \mathbb{R}$ is *admissible* if:

$$
\begin{aligned}
0 \le \sigma_a(v,z), \sigma_p(v,z), &\quad \text{for all} \quad v,z \\
\sigma(v,z) \ge \sigma_a(v,z) + \sigma_p(v,z), &\quad \text{for all} \quad v,z \\
\sigma_a(v,z) \text{ is increasing in } z, &\quad \text{for all} \quad v \\
\sigma_p(v,z) \text{ is increasing in } v, &\quad \text{for all} \quad z
\end{aligned}
$$

Division of Surplus: Noncompetitive Markets

Noncompetitive markets: an exogenous (market-wide) division rule: NTU.

Definition

Given $\sigma_a, \sigma_p, \Psi_m$, an *m equilibrium* is a matching $\left(\nu, \sigma_{a,m}, \sigma_{p,m}\right)$ such that

$$\sigma_{a,m}(\nu(z)) = \sigma_a(\nu(z), z), \quad \text{for all} \quad z \qquad (\text{Feas}_v)$$

$$\sigma_{p,m}(z) = \sigma_a(\nu(z), z), \quad \text{for all} \quad z \qquad (\text{Feas}_z)$$

$$\sigma_{p,m}(z) < \sigma_p(v, z) \Rightarrow \sigma_{a,m}(v) \geq \sigma_a(v, z), \quad \text{for all} \quad z, v \qquad (\text{Stab})$$

Lemma

Given admissible σ_a, σ_p, for all Ψ_m, PAM is supported by an m equilibrium.
Under PAM in m, an agent with quality v receives $\sigma_{a,m}(v) := \sigma_a\left(v, G_m^{-1}(F_m(v))\right)$.

Definition

An equilibrium of the game is a strategy $\Phi : \mathbb{R} \to \Delta(\mathcal{M})$, distributions of agents $(\Psi_1, ..., \Psi_M)$, and values for agents $\left(\sigma_{a,m}(v)\right)_{m \in \mathcal{M}}$ such that

(i) For all m and v, $\sigma_{a,m}(v)$ represent equilibrium payoffs in the market-subgame given Ψ_m;

(ii) If $\Phi_m(x) > 0$, then $m \in \operatorname{argmax}_{\tilde{m}} \mathbb{E}_x \left[\sigma_{a,\tilde{m}}(F^{-1}(r))\right]$; and

(iii) $d\Psi_m(r) = \int \Phi_m(x) \cdot \mu(r, dx)$.

Main Result

Within-Sector Efficiency

Regardless of competition in the market, the resulting matching is efficient:

Lemma

Given an allocation of agents Ψ_m, there is a unique competitive market subgame equilibrium. It is efficient. If the market is not competitive, there is an efficient stable matching.

- This allows me to focus exclusively on Ψ_m.

Definition

Given Ψ_m, *aggregate surplus* in market m is: $W_m(\Psi_m) \coloneqq \int_0^1 \sigma\left(F_m^{-1}(r), G_m^{-1}(r)\right) dr$.

Given agents' information, the feasible $\Psi = (\Psi_1, ..., \Psi_M)$ are

$$\mathcal{F}(\mu) \coloneqq \left\{ \Psi : \exists \Phi : \mathbb{R} \to \Delta(\mathcal{M}) \text{ measurable, } d\Psi_m(r) = \int \Phi_m(x) \cdot \mu(r, dx) \right\}$$

Allocative (In)Efficiency

Definition

Given Ψ_m, *aggregate surplus* in market m is: $W_m(\Psi_m) \coloneqq \int_0^1 \sigma\left(F_m^{-1}(r), G_m^{-1}(r)\right) dr$.

Given agents' information, the feasible $\Psi = (\Psi_1, ..., \Psi_M)$ are

$$\mathcal{F}(\mu) \coloneqq \left\{\Psi : \exists \Phi : \mathbb{R} \to \Delta(\mathcal{M}) \text{ measurable, } d\Psi_m(r) = \int \Phi_m(x) \cdot \mu(r, dx)\right\}$$

Definition

An allocation of producers $\Psi^* = \left(\Psi_1^*, ..., \Psi_M^*\right)$ is *allocatively efficient* (*given* μ) if

$$\Psi^* \in \underset{\Psi \in \mathcal{F}(\mu)}{\operatorname{argmax}} \sum_{m \in \mathcal{S}} W_m(\Psi_m)$$

I say that Ψ^* is *allocatively inefficient* if it is not allocatively efficient.

Main Result: Both Frictions $\iff$ Allocative Inefficiency

Theorem

Fix σ, admissible σ_a, σ_p, and μ.

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Fix σ, admissible σ_a, σ_p, and μ.

(i) If agents have perfect information, $\mu = \mu_C$, the unique equilibrium is efficient;

(ii) If markets are competitive, the unique equilibrium is efficient; but

(iii) If agents have incomplete information $\mu \neq \mu_C$ and markets are not competitive, generically* the unique equilibrium is inefficient.

*Genericity is with respect to the distributions of agents and positions (F, G).

Proof Sketch: (i) Complete information implies efficiency

If $\mu = \mu_C$, then any allocation Ψ is feasible.

- Efficiency is equivalent to, *across sectors*, PAM being satisfied.

Noncompetitive markets:

- The best agents will go to the sectors with the best positions $\Rightarrow$ PAM.

Competitive markets:

- Less obvious: divisions $\sigma_{a,m}$ could differ across sectors.
- In equilibrium, competitive forces are the same across sectors $\Rightarrow$ best agents go to the sectors with the best positions.

Proof Sketch: (ii) Competition implies efficiency

Lemma

$W_m(\Psi_m)$ is strictly concave, as a function of Ψ_m.

Corollary

There exists a unique efficient Ψ. It is identified from first order conditions.

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Corollary

There exists a unique efficient Ψ. It is identified from first order conditions.

The bulk of the proof is showing that, for all allocations Ψ_m,

$$\frac{d}{d\epsilon} W_m\left(\Psi_m + \epsilon\delta_{F(v)}\right)\Big|_{\epsilon=0} = \sigma_{a,m}(v;\Psi_m)$$

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$$\frac{d}{d\epsilon} W_m\left(\Psi_m + \epsilon\mu_x\right)\Big|_{\epsilon=0} = \int \sigma_{a,m}(F^{-1}(r);\Psi_m)\mu(r, dx)$$

At an optimal allocation types x go to the sectors with largest benefit. These are exactly those with the largest competitive equilibrium surplus divisions.

Proof Sketch: (iii) Noncompetitive Efficiency is Knife-Edge

Suppose that $\mu \neq \mu_C$.

- In competitive equilibrium, divisions rely on the *entire* distribution of assignments in a sector.
- In a noncompetitive equilibrium, agent v's payoff depends only on his match.

Proof Sketch: (iii) Noncompetitive Efficiency is Knife-Edge

Suppose that $\mu \neq \mu_C$.

- In competitive equilibrium, divisions rely on the *entire* distribution of assignments in a sector.
- In a noncompetitive equilibrium, agent v's payoff depends only on his match.

First, perturb G so that the competitive equilibrium is not degenerate. Then, there is some type x^* indifferent between m, m'.

- Perturb G to keep *all* agents' competitive payoffs in each market m the same, but
- change the noncompetitive payoff so that type x^* strictly prefers m to m'.
- Given μ, may need to also perturb F to ensure that this is possible.

Applications

Comparative Statics: Concentration of Environments

Comparative statics depend on the specific application.

Definition

- An environment: $E = \left[\left(\sigma, \sigma_a, \sigma_p\right), \left(\mu, F\right)\right]$.
- A complete environment: (E, G) an environment E with $G = (G_1, ..., G_M)$.

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First, study how *concentrated* are environments:

Definition

- Given (E, G), let $A_{E,G} \coloneqq \{s \in \mathcal{S} : \Psi_m \neq 0, \text{ some eq'm}\}$ denote its *active sectors*.
- E is *more concentrated* than E' if for all G, $A_{E,G} \subseteq A_{E',G}$.

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Definition

- Given (E, G), $x \in \mathbb{R}$, let $A_{E,G}(x) \coloneqq \{s \in \mathcal{S} : \Phi_m(x) > 0, \text{ some eq'm}\}$ denote x's *active sectors*.
- Fix E, E' with $\mu_E = \mu_{E'}$. $x \in \mathbb{R}$:
  θ *concentrates more* in E than E' if, for all G, $A_{E,G}(x) \subseteq A_{E',G}(x)$.

Comparative Statics: Concentration of Sectors

Second, study concentration *within* sectors:

Definition

G is *more concentrated* than G' if G' is a mean-preserving spread of G.

When agents have no private information ($\mu = \mu_N$), the unique equilibrium just specifies the mass of agents Φ_m in each market m.

Definition

Fix $E = \left[\left(\sigma, \sigma_a, \sigma_p\right), \left(\mu_N, F\right)\right]$. Concentrated sectors are *preferred* in E if, for all G_1, G_1' with G_1 more concentrated than G_1' and all G,

$$\Phi_1\Big[\left(E, \left(G_1, G\right)\right)\Big] \geq \Phi_1\Big[\left(E, \left(G_1', G\right)\right)\Big]$$

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Definition

Fix $E = \left[\left(\sigma, \sigma_a, \sigma_p\right), \left(\mu_N, F\right)\right]$. Concentrated sectors are *eventually preferred* in E if, for all G_1, G_1' with G_1 more concentrated than G_1' there exists $\overline{z}, \delta > 0$ such that for all G with $G_m(\overline{z}) \leq \delta$ for all $G_m \in G$,

$$\Phi_1\Big[\left(E, \left(G_1, G\right)\right)\Big] \geq \Phi_1\Big[\left(E, \left(G_1', G\right)\right)\Big]$$

Worker Sorting

Worker Sorting: Setup

A one-to-one mapping to our framework:

- Agents: Workers, with productivity $v_i \sim F$.
- Positions: Firms, with idiosyncratic productivity $z_{j_m} \sim G_m$ in sector m.

Together, a v worker and z firm produce output $\sigma(v, z) = v \cdot z$.

- If wage frictions exist, a worker is paid proportionately to their output: $\sigma_a(v, z) = \alpha\sigma(v, z)$.

Assumption

F has increasing hazard rates: $\frac{f(v)}{1-F(v)}$ is increasing.

Worker Sorting: Concentration with Grade Inflation

Definition

μ features "grade inflation" if there exists an $\bar{r}$ such that

$$\mu_r = \begin{cases} \delta_r & \text{if} \quad r < \bar{r}, \\ U([\bar{r}, 1]) & \text{if} \quad r \geq \bar{r} \end{cases}$$

Proposition

Given a μ with grade inflation, top workers concentrate more in the noncompetitive equilibrium than in the efficient allocation.

- In noncompetitive markets, top workers do not internalize their displacement of other workers.
- This leads them to compete too heavily for positions in attractive sectors.

Worker Sorting: Workers prefer riskier sectors than is optimal

Now, workers have no information: $\mu = \mu_N$.

Proposition

(i) If markets are noncompetitive, less concentrated sectors are preferred; but

(ii) If markets are competitive, concentrated sectors are eventually preferred.

When markets are noncompetitive, workers prefer *risky* sectors: those with some very attractive positions and many unattractive positions. Instead, a planner prefers many similar positions, since returns are high.

Innovation

Innovation: Model I

There is a unit mass of researchers $i \in [0, 1]$.

- Researcher i achieves an innovation at rate λ, independently;
- Researcher i chooses a discrete "sub-field" in which to work.

There are $\mathcal{M}$ *sub-fields*.

- In sub-field m there is a distribution of innovations of varying quality $z \sim G_m$.
- Think of as the value of the innovation.

Time is discounted by society at rate r, and by researchers at rate δ, with $\delta \geq r$.

Innovation: Model II

Researchers commit to a sub-field, and then compete with other researchers in their sub-field to achieve innovations.

- The first researcher to achieve an innovation z at time t receives $ze^{-\delta t}$. This captures the benefit from patenting.
- The social benefit is ze^{-rt}.

Consider two regimes. After his first achieved innovation, researchers either

- Exit (develop a research agenda); or
- Remain (sequential research).

Innovation: Mapping to Framework

Agents: Researchers, with qualities e^{-rt}.

Positions: Innovations, with value z.

Surplus generation function $\sigma(e^{-rt}, z) = z \cdot e^{-rt}$, with $\sigma_a(e^{-rt}, z) = z \cdot e^{-\delta t}$.

Information $\mu = \mu_N$: arrivals are random.

The last piece is the distribution F. This depends on the regime:

Lemma

- In *Exit*: $e^{-rt} \sim \text{Beta}\left(\frac{\lambda}{r}, 1\right)$;
- In *Remain*: e^{-rt} follows an improper distribution on $(0, 1]$, with density $\frac{\lambda}{r} \cdot \frac{1}{x}$.

In "Remain", all projects are eventually completed, so the distribution is improper.

Innovation: Concentration Results

Proposition

Suppose that $\delta \geq r$. In both regimes:

(i) Researchers are more concentrated than is socially optimal;

(ii) As the rate of discovery λ increases, concentration decreases; and

(iii) As researchers become more impatient (δ increases), concentration increases.

- As in the worker application, market frictions cause agents to concentrate too heavily.
- Impatience (e.g., short funding windows) exacerbates this problem.

Innovation: Researchers prefer risky fields

Proposition

Let $\delta = r$. In both regimes,

(i) researchers prefer less concentrated (risky) sub-fields;

(ii) if markets were competitive and $\frac{\lambda}{r} \geq 1$, then concentrated subfields are eventually preferred; but

(iii) If $\frac{\lambda}{r} < 1$, concentrated subfields are not eventually preferred.

The "speed" of innovation matters in competitive markets:

- If $\frac{\lambda}{r} < 1$, innovations are "slow", and so later innovations hold little value.
- The "most preferred" sectors (subject to average innovation) will equalize value across the first $\frac{\lambda}{r}$ mass of projects, and set the value of others to zero.

Innovation: Comparison of Regimes

Proposition

(i) Researchers concentrate more in *Exit* than in *Remain*; but

(ii) In the efficient allocations, it is ambiguous there is more concentration in *Exit* than in *Remain*.

- In *Exit*, competition forces accumulate more slowly: if a researcher is unlucky, competitors will have already left.
- In an efficient market there are two forces. In *Remain*
  - more projects are solved per researcher $\Rightarrow$ spread out; but
  - an additional researcher can finish multiple projects $\Rightarrow$ concentrate.

  Which force wins depends on G.

Content Production

Content Production: Model I

There is a unit mass of producers.

- Each producer produces a good of random value $v \sim F$.

There are M "genres" ("styles") of goods.

- Producers choose in which genre they produce, but do not know their quality ex-ante: $\mu = \mu_N$: "Nobody Knows Anything."

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There is a unit mass of heterogeneous consumers with types $\xi \in \Xi$.

- Type $\xi = (t_1, ..., t_M)$ can consume up to $t_m \in [0, 1]$ mass of content in genre m.
- Let $T_m(t)$ is the mass of consumers willing to consume at most t units of content in genre m.
  - Assume that T_m is strictly increasing and continuous (may have an atom at zero).

Content Production: Model II

Consumer of type $\xi = (t_1, ..., t_M)$ receives utility proportional to the quality of goods that he consumes up to his capacity constraint, net their cost.

- He consumes the t_m goods in genre m that give the highest net utility $v - p_m(v)$.

Producers maximize profits, anticipating prices and demand in each genre.

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Today: consider two pricing cases.

- Competitive Pricing: After qualities are realized, producers set prices in each genre to maximize demand $p_m(v)$;
- "Streaming": Consumers do not pay per good, producers aim to maximize demand.

Content Production: Mapping to Framework

Agents: Producers with quality v.

Positions: Consumer demand, with quantity z.

- The distribution of demand in genre m is $G_m(z) \coloneqq T_m^{-1}(z)$.

Surplus generation function: $\sigma(v, z) = v \cdot z$.

Surplus division $\sigma_a(v, z)$ depends on the pricing scheme:

- Competitive pricing: determined in equilibrium.
- *Streaming* pricing: $\sigma_a(v, z) = z$.

Information $\mu = \mu_N$: producers do not know their quality.

Content Production: The mapping works.

Lemma

There exists a unique equilibrium of the competitive pricing sector sub-game that corresponds to the unique competitive equilibrium in the general model.

This isn't obvious.

- Producers are the only ones able to make offers, and they must choose a per-unit price.
- Consumers also decide consumption not at the z-level, but individually.
- The assumption that T_m has no atoms (except at zero) is important.

Content Production: Concentration depends on tail-weight

Definition

$\frac{f(v)}{1-F(v)}$ is the hazard rate of F.

Proposition

(i) If F has *increasing* hazard rates, the *Streaming* equilibrium is more concentrated than the efficient allocation; but

(ii) If F has *decreasing* hazard rates, the *Streaming* equilibrium is less concentrated than the efficient allocation.

Content Production: Homogeneous vs. Heterogeneous Consumer Demand

Lemma

If T is more concentrated than T', then the induced G is *less* concentrated than G'.

Comparative statics on T correspond to the "reverse" comparative statics on G.

Proposition

- In the *Streaming* setting, producers prefer homogeneous-demand genres; but
- If F has IHR, efficiently, genres with more heterogeneous consumer demand are eventually preferred.

Content Production: Other Results

Other comparative statics:

- Quality-dependent pricing.
- Quantity-dependent pricing.
- Changes to the value distribution.

I also study how other market structures interact with these inefficiencies:

- A monopolist subject to pricing constraints.
- A monopolist using genre-specific subscriptions.
- A “recommendation system” that connects producers to consumers.

Future Directions

- Other types of comparative statics , possibly for general production functions σ and information structures μ.
- General mechanisms for determining surplus divisions.
  - Currently, just the extremes of competitive markets and deterministic divisions. Would like to study intermediate cases.
  - Premuneration values? Incomplete information within sub-markets?
- Other possible applications?

Related Literature

Matching markets, competitive equilibrium:

- Koopmans and Beckman (1957), Terviö (2008), Kolotilin and Wolitsky (2026)

Worker Sorting:

- Lindenlaub (2017), Lise and Postel-Vinay (2020), Alon and Fershtman (2025)

Innovation:

- Bhattacharya and Mookherjee (1986), Bryan and Lemus (2017), Hopenhayn and Squintani (2021)

Content Production:

- Aguiar and Waldfogel (2018), Gambato (2025), Chen, Li, and Preuss (2025)

Conclusion

I develop a model to understand how information and competition frictions affect *horizontal* investment decisions.

- I show that the uncertainty, together with a lack of competitive pricing, leads to inefficient sorting.

I study three different applications of the framework:

(i) **Worker sorting**: Wage frictions lead to too much concentration for workers.

(ii) **Innovation**: Researchers concentrate and over-value "risky" fields.

(iii) **Content markets**: Direction of inefficiency depends on the quality distribution's tails.